

1. General Description

The EM74HC7002; EM74HCT7002 is a quad 2-input NOR gate with Schmitt-trigger inputs. Schmitt trigger inputs transform slowly changing input signals into sharply defined jitter-free output signals.

2. Features and Benefits

- Wide supply voltage range from 2.0 V to 6.0 V
- High noise immunity
- CMOS low power dissipation
- Latch-up performance exceeds 250 mA
- Unlimited rise and fall times
- Complies with JEDEC standards:
 - JESD8C (2.7 V to 3.6 V)
 - JESD7A (2.0 V to 6.0 V)
- ESD protection:
 - HBM ANSI/ESDA/JEDEC JS-001 Class 2 exceeds 3500 V
 - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V
- Multiple package options

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Quad 2-input NOR Schmitt trigger

3. Ordering Information

Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EM74HC7002D	HC7002□ XXXWW	SOP-14L	plastic small outline package; 14 leads; body width 3.9 mm	3000
EM74HCT7002D	HCT7002□ XXXWW			
EM74HC7002PW	HC7002□ XXXWW	TSSOP-14L	plastic thin shrink small outline package; 14 leads; body width 4.4 mm	3000
EM74HCT7002PW	HCT7002□ XXXWW			

MARKING INFORMATION

NOTE: HC7002□ = Serial number and Internal code; XXXWW = Vendor Code and Date Code.

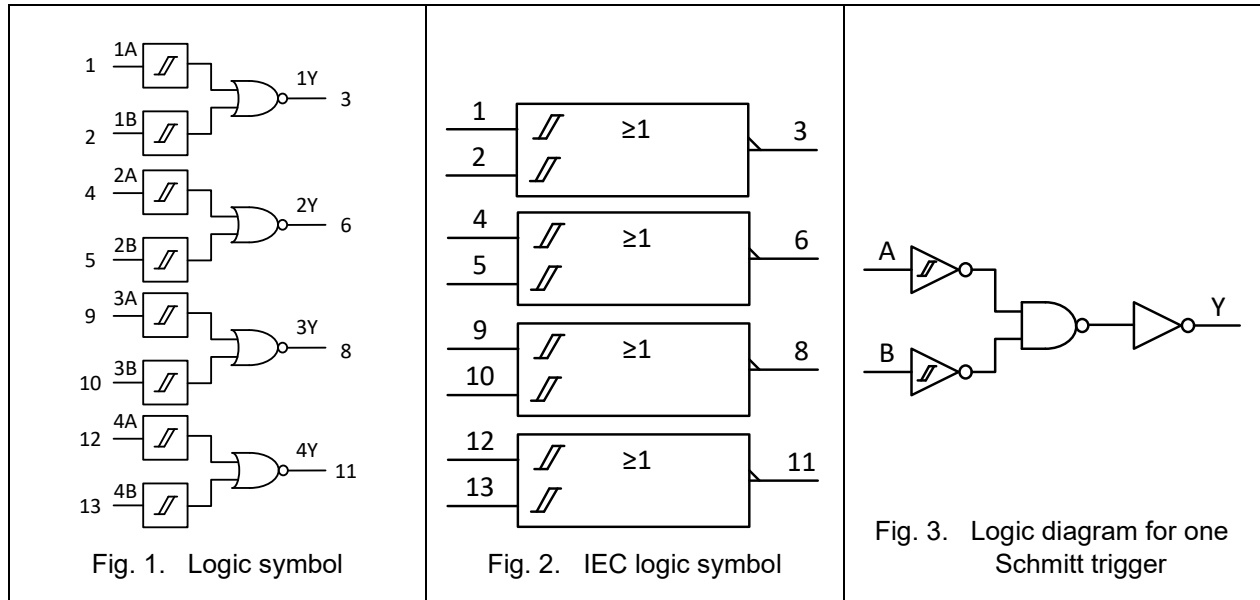
HC7002 □

Internal Code
Serial number

X YY WW

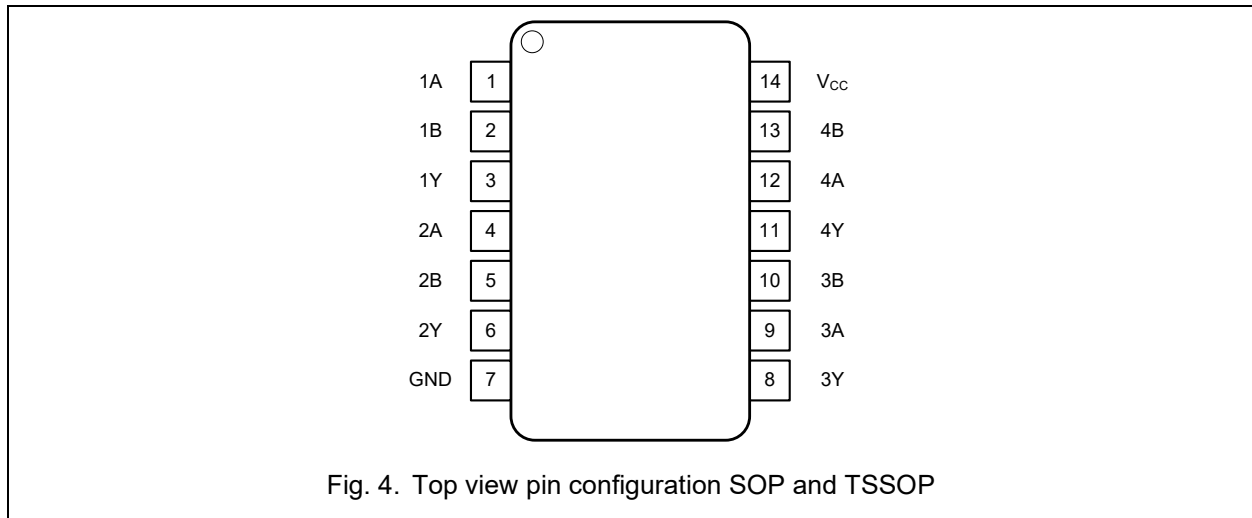
Date Code -Week
Date Code -Year
Vendor Code

4. Function Diagram



5. Pinning Information

5.1. Pinning



5.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
1A, 2A, 3A, 4A	1, 4, 9, 12	Data input
1B, 2B, 3B, 4B	2, 5, 10, 13	Data input
1Y, 2Y, 3Y, 4Y	3, 6, 8, 11	Data output
GND	7	Ground (0V)
V _{CC}	14	Supply voltage

6. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level.

Control		Output
nA	nB	nY
L	L	H
L	H	L
H	L	L
H	H	L

7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V_{CC}	supply voltage		-0.5	7.0	V
I_{IK}	input clamping current	$V_I < -0.5\text{ V}$ or $V_I > V_{CC} + 0.5\text{ V}$ [1]		± 20	mA
I_{OK}	output clamping current	$V_O < -0.5\text{ V}$ or $V_O > V_{CC} + 0.5\text{ V}$ [1]		± 20	mA
I_O	output current	$V_O = -0.5\text{ V}$ to $(V_{CC} + 0.5\text{ V})$		± 25	mA
I_{CC}	supply current			50	mA
I_{GND}	ground current		-50		mA
P_{tot}	total power dissipation			500	mW
T_{stg}	storage temperature		-65	150	°C

[1] The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Conditions	EM74HC7002			EM74HCT7002			Unit
			Min	Typ	Max	Min	Typ	Max	
V_{CC}	supply voltage		2.0	5.0	6.0	4.5	5.0	5.5	V
V_I	input voltage		0		V_{CC}	0		V_{CC}	V
V_O	output voltage		0		V_{CC}	0		V_{CC}	V
T_{amb}	ambient temperature		-40		125	-40		125	°C

9.Static Characteristics

Table 6. Static characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V). Typical values measured at $T_{amb} = 25^{\circ}\text{C}$ (unless otherwise noted).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
EM74HC7002								
V _{OH}	HIGH-level output voltage	V _I = V _{T+} or V _{T-}						
		I _O = -20 μA; V _{CC} = 2.0 V	1.9			1.9		V
		I _O = -20 μA; V _{CC} = 4.5 V	4.4			4.4		V
		I _O = -20 μA; V _{CC} = 6.0 V	5.9			5.9		V
		I _O = -4.0 mA; V _{CC} = 4.5 V	3.84			3.7		V
		I _O = -5.2 mA; V _{CC} = 6.0 V	5.34			5.2		V
V _{OL}	LOW-level output voltage	V _I = V _{T+} or V _{T-}						
		I _O = 20 μA; V _{CC} = 2.0 V			0.1		0.1	V
		I _O = 20 μA; V _{CC} = 4.5 V			0.1		0.1	V
		I _O = 20 μA; V _{CC} = 6.0 V			0.1		0.1	V
		I _O = 4.0 mA; V _{CC} = 4.5 V			0.33		0.4	V
		I _O = 5.2 mA; V _{CC} = 6.0 V			0.33		0.4	V
I _I	input leakage current	V _I = V _{CC} or GND ; V _{CC} = 6.0 V			±1		±1	μA
I _{CC}	supply current	V _I = V _{CC} or GND ; I _O = 0 A ; V _{CC} = 6.0 V			20		40	μA
C _I	input capacitance			7				pF

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Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
EM74HCT7002								
V _{OH}	HIGH-level output voltage	V _I = V _{T+} or V _{T-} ; V _{CC} = 4.5 V						
		I _O = -20 μA;	4.4			4.4		V
		I _O = -4.0 mA;	3.84			3.7		V
V _{OL}	LOW-level output voltage	V _I = V _{T+} or V _{T-} ; V _{CC} = 4.5 V						
		I _O = 20 μA;			0.1		0.1	V
		I _O = 4.0 mA;			0.33		0.4	V
I _I	input leakage current	V _I = V _{CC} or GND ; V _{CC} = 5.5 V			±1		±1	μA
I _{CC}	supply current	V _I = V _{CC} or GND; I _O = 0 A; V _{CC} = 5.5 V			20		40	μA
ΔI _{CC}	additional supply current	per pin ; V _I = V _{CC} - 2.1 V; I _O = 0 A; other inputs at V _{CC} or GND; V _{CC} = 4.5 V to 5.5 V			1.55		1.85	mA
C _I	input capacitance			10				pF

10. Dynamic Characteristics

Table 7. Dynamic characteristics

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 6. Typical values measured at $T_{amb} = 25^{\circ}\text{C}$ (unless otherwise noted).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
EM74HC7002								
t _{pd}	propagation delay	nA, nB to nY; see Fig. 5 [1]						
		V _{CC} = 2.0 V			35		40	ns
		V _{CC} = 4.5 V			20		25	ns
		V _{CC} = 6.0 V			15		20	ns
t _t	transition time	see Fig. 5 [2]						
		V _{CC} = 2.0 V			9		11	ns
		V _{CC} = 4.5 V			6		8	ns
		V _{CC} = 6.0 V			4		5	ns
C _{PD}	power dissipation capacitance	C _L = 15 pF; f = 1 MHz; V _I = GND to V _{CC} [3]		18				pF
EM74HCT7002								
t _{pd}	propagation delay	nA, nB to nY; V _{CC} = 4.5 V; see Fig. 5 [1]			16		19	ns
t _t	transition time	V _{CC} = 4.5 V; see Fig. 5 [2]			6		8	ns
C _{PD}	power dissipation capacitance	C _L = 15 pF; f = 1 MHz; V _I = GND to (V _{CC} -1.5 V) [3]		37				pF

[1] t_{pd} is the same as t_{PHL} and t_{PLH} .

[2] t_t is the same as t_{THL} and t_{TLH} .

[3] C_{PD} is used to determine the dynamic power dissipation (P_D in μW).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \Sigma(C_L \times V_{CC}^2 \times f_o)$ where:

f_i = input frequency in MHz;

f_o = output frequency in MHz;

C_L = output load capacitance in pF;

V_{CC} = supply voltage in V;

N = number of inputs switching;

$\Sigma(C_L \times V_{CC}^2 \times f_o)$ = sum of outputs.

10.1. Waveforms and test circuit

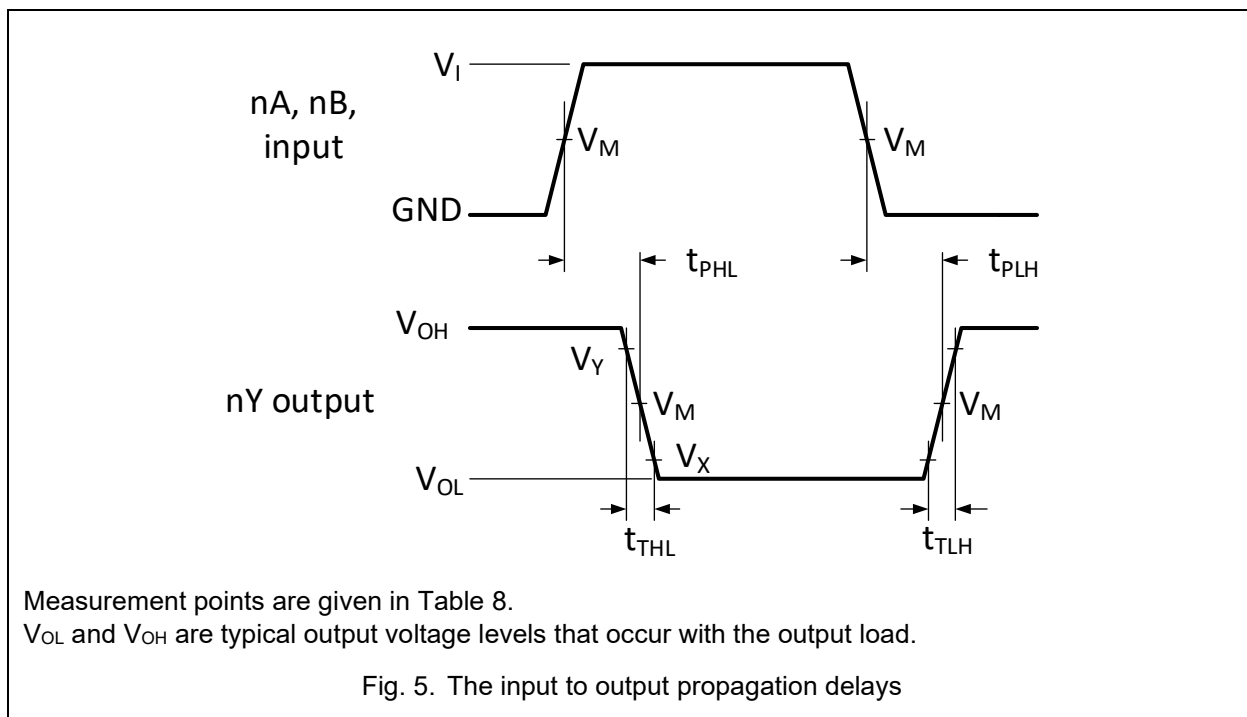
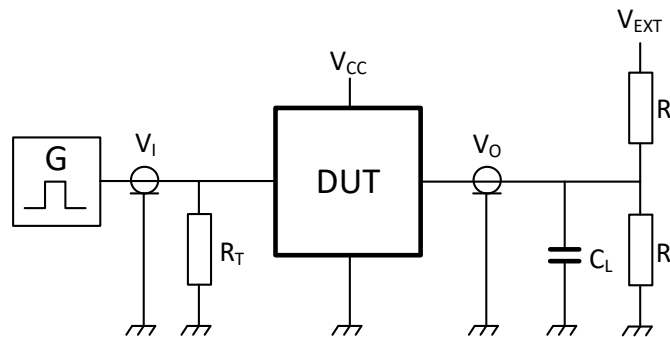
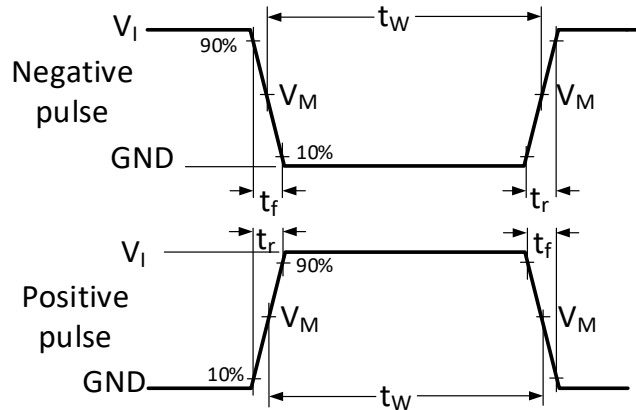


Table 8. Measurement points

Type	Input	Output		
	V_M	V_M	V_X	V_Y
EM74HC7002	$0.5V_{CC}$	$0.5V_{CC}$	$0.1V_{CC}$	$0.9V_{CC}$
EM74HCT7002	1.3 V	1.3 V	$0.1V_{CC}$	$0.9V_{CC}$

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Test data is given in Table 9.

Definitions for test circuit:

R_L = Load resistance.

C_L = Load capacitance including jig and probe capacitance.

R_T = Termination resistance should be equal to the output impedance Z_o of the pulse generator.

V_{EXT} = External voltage for measuring switching times.

Fig. 6. Test circuit for measuring switching times

Table 9. Test data

Type	Input		Load		V_{EXT}
	V_I	$t_r = t_f$	C_L	R_L	t_{PLH}, t_{PHL}
EM74HC7002	V_{CC}	$\leq 2.5 \text{ ns}$	15 pF	500 Ω	open
EM74HCT7002	3 V	$\leq 2.5 \text{ ns}$	15 pF	500 Ω	open

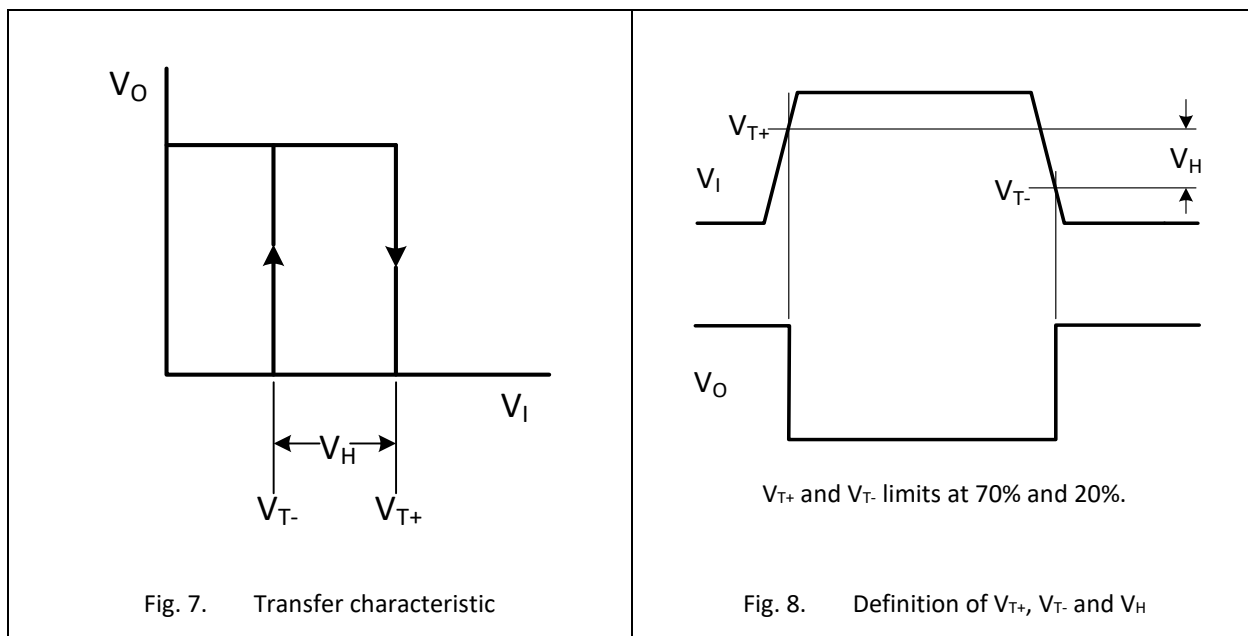
11. Transfer Characteristics

Table 10. Transfer characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V). Typical values are measured at $T_{amb} = 25^{\circ}\text{C}$.

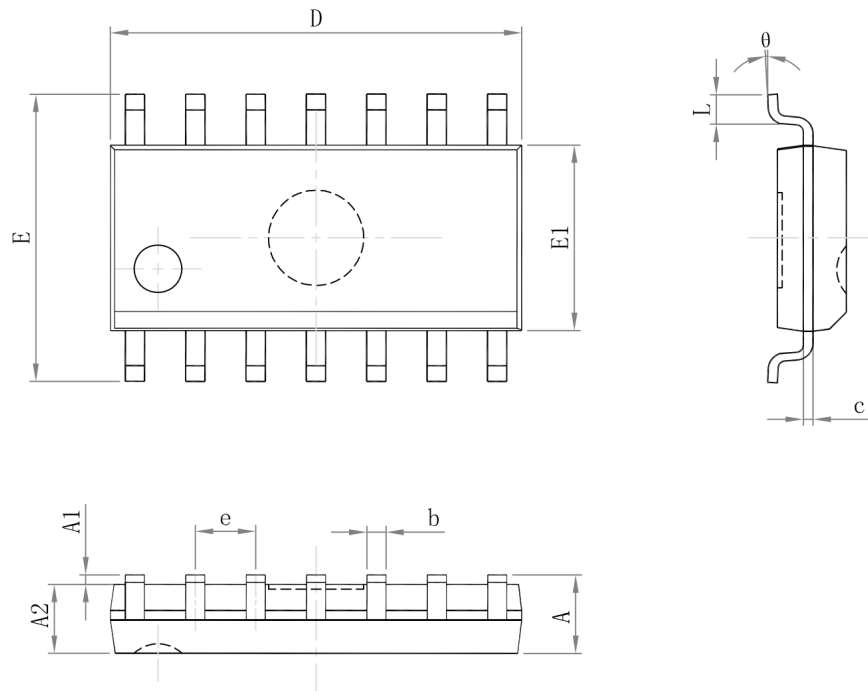
Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
EM74HC7002								
V _{T+}	positive-going threshold voltage	see Fig. 7 and Fig. 8						
		V _{CC} = 2.0 V	0.7	1.21	1.5	0.7	1.5	V
		V _{CC} = 4.5 V	1.7	2.39	3.15	1.7	3.15	V
		V _{CC} = 6.0 V	2.1	3.05	4.2	2.1	4.2	V
V _{T-}	negative-going threshold voltage	see Fig. 7 and Fig. 8						
		V _{CC} = 2.0 V	0.3	0.6	0.9	0.3	0.9	V
		V _{CC} = 4.5 V	0.9	1.41	2.0	0.9	2.0	V
		V _{CC} = 6.0 V	1.2	1.9	2.6	1.2	2.6	V
V _H	hysteresis voltage	see Fig. 7 and Fig. 8						
		V _{CC} = 2.0 V	0.2	0.61	1.0	0.2	1.0	V
		V _{CC} = 4.5 V	0.4	0.97	1.4	0.4	1.4	V
		V _{CC} = 6.0 V	0.6	1.16	1.6	0.6	1.6	V
EM74HCT7002								
V _{T+}	positive-going threshold voltage	see Fig. 7 and Fig. 8						
		V _{CC} = 4.5 V	1.2	1.45	1.9	1.2	1.9	V
		V _{CC} = 5.5 V	1.4	1.6	2.1	1.4	2.1	V
V _{T-}	negative-going threshold voltage	see Fig. 7 and Fig. 8						
		V _{CC} = 4.5 V	0.5	0.88	1.2	0.5	1.2	V
		V _{CC} = 5.5 V	0.6	1.05	1.4	0.6	1.4	V
V _H	hysteresis voltage	see Fig. 7 and Fig. 8						
		V _{CC} = 4.5 V	0.4	0.57		0.4		V
		V _{CC} = 5.5 V	0.4	0.55		0.4		V

11.1. Waveforms transfer characteristics

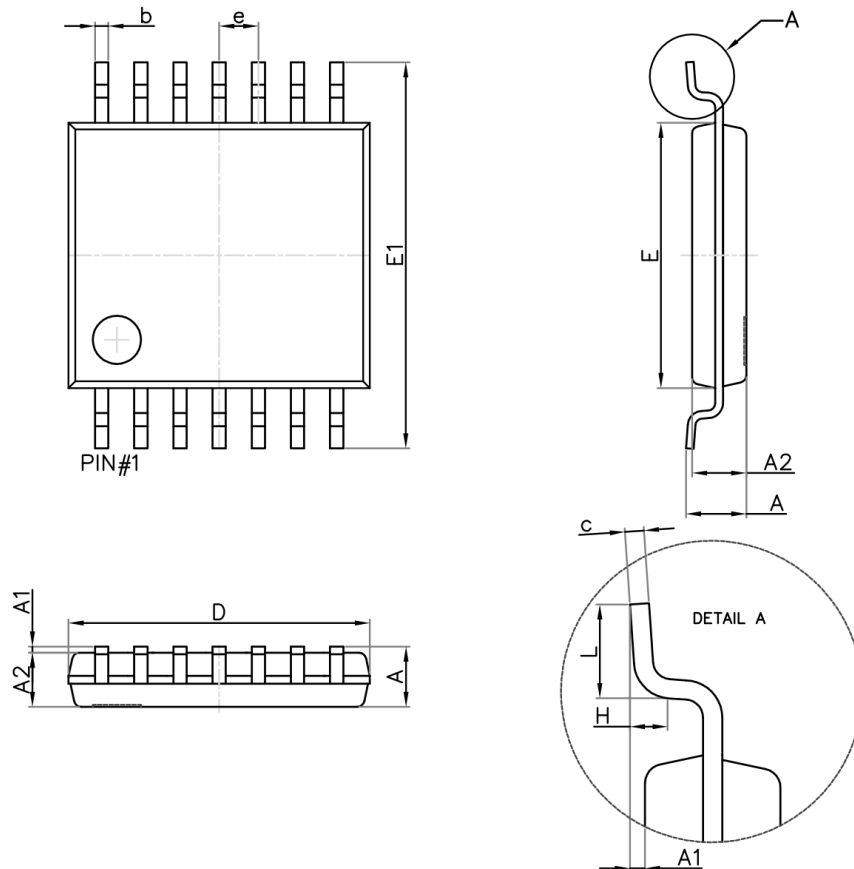


12. Package Outline

SOP-14L



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	—	1.750	—	0.069
A1	0.100	0.250	0.004	0.010
A2	1.250	—	0.049	—
b	0.310	0.510	0.012	0.020
c	0.100	0.250	0.004	0.010
D	8.450	8.850	0.333	0.348
E	5.800	6.200	0.228	0.244
E1	3.800	4.000	0.150	0.157
e	1.270(BSC)		0.050(BSC)	
L	0.400	1.270	0.016	0.050
theta	0°	8°	0°	8°

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TSSOP-14L


Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
D	4.900	5.100	0.193	0.201
E	4.300	4.500	0.169	0.177
b	0.190	0.300	0.007	0.012
c	0.090	0.200	0.004	0.008
E1	6.250	6.550	0.246	0.258
A	—	1.200	—	0.047
A2	0.800	1.000	0.031	0.039
A1	0.050	0.150	0.002	0.006
e	0.65 (BSC)		0.026 (BSC)	
L	0.500	0.700	0.020	0.028
H	0.25(TYP)		0.01(TYP)	
θ	1°	7°	1°	7°

13. Tape and Reel Information

13.1. Carrier tape dimensions

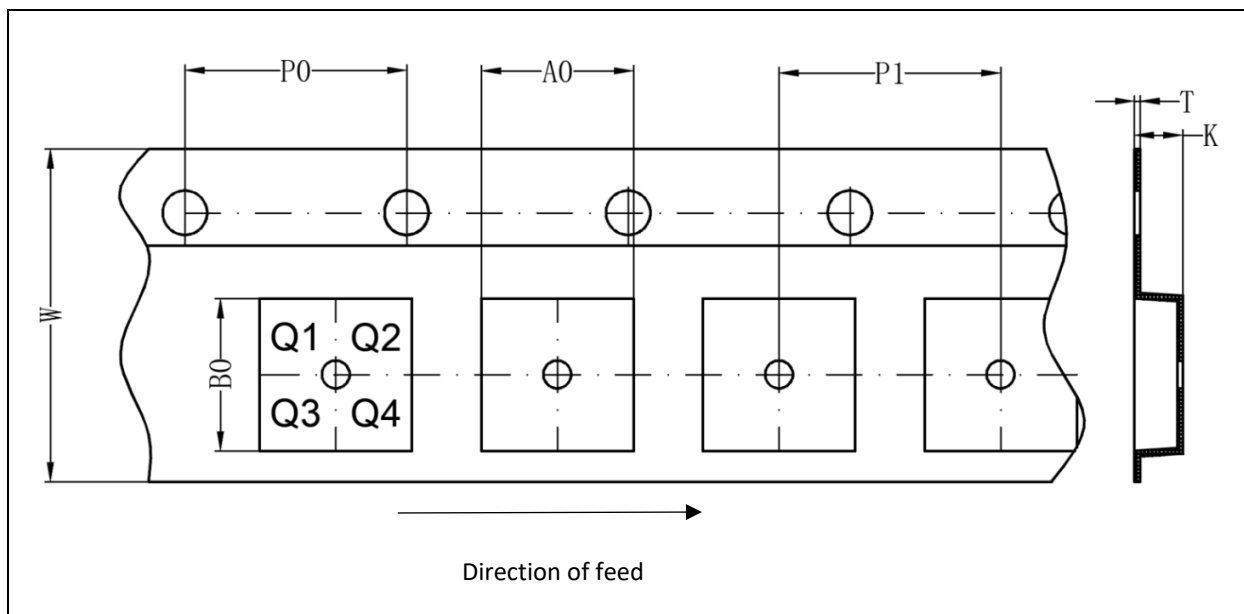


Table 11. Carrier tape dimensions

Package version	A0(mm)	B0(mm)	K0(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
SOP-14L	6.35	9.1	1.9	0.22	8	16	4	Q1
TSSOP-14L	6.7	5.45	1.6	0.25	8	12	4	Q1

13.2. Reel and box dimensions

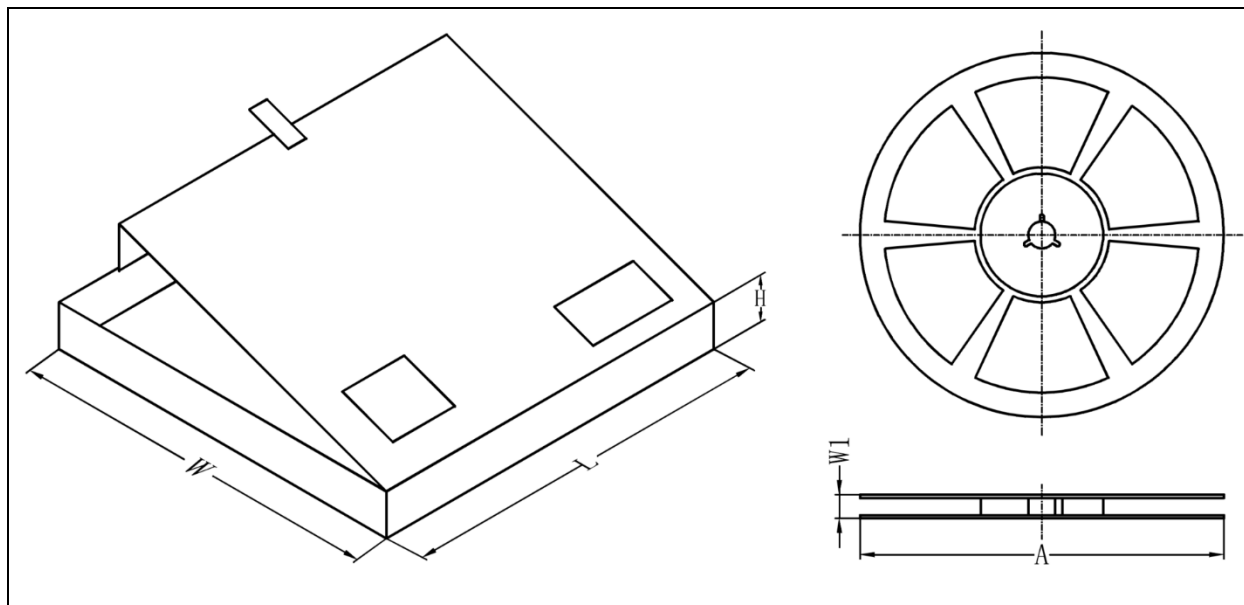


Table 12. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Reel Width W1 (mm)	SPQ (pcs) [1]	Reels per box	Outer box dimensions L×W×H(mm) [2]
SOP-14L	D	330	22.4	3000	1	358x340x50
TSSOP-14L	PW	330	18.4	3000	1	358x340x50

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

14. Abbreviations

Table 13. Abbreviations

Acronym	Description
CMOS	Complementary Metal-Oxide Semiconductor
DUT	Device Under Test
ESD	ElectroStatic Discharge
HBM	Human Body Model
CDM	Charged Device Model
TTL	Transistor-Transistor Logic

15. Revision History

Table 14. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EM74HC_HCT7002 Rev. 1.1	Dec 15, 2025	Product datasheet		EM74HC_HCT7002 Rev. 1.0
Modifications:	- Table 1: topside marking information updated. - Section 13 added Tape and Reel Information.			
EM74HC_HCT7002 Rev. 1.0	Aug 30, 2024	Product datasheet		